WORKED SOLUTION

STEP II 2004, Question 8

Pure • Comparing Differential Equations

STEP II 2004, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q8

The key idea

While $0 \leq x < 1$ the right-hand side is positive, so x increases with t and we may turn the equation over: t becomes a function of x, namely the time taken to climb from 0 to height x. A bigger gradient at every height means every height is reached sooner, so the given inequalities between the right-hand sides immediately order the three graphs, one above the next. Whether the climb to $x = 1$ takes finite or infinite time is then a question about whether an improper integral converges.

Solution

Throughout, write $x_n(t)$ for the solution with exponent n. At $t = 0$ we have $x = 0$ and gradient $(1-0)^{1/n} = 1$ for every n, so all the curves leave the origin along the line $x = t$. While $0 \leq x < 1$ the gradient is strictly positive, so x is increasing and

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{(1-x^n)^{1/n}}, \qquad \text{so} \qquad t_n(x) = \int_0^x \frac{\mathrm{d}\xi}{(1-\xi^n)^{1/n}}$$

is the time at which the solution reaches height x, for $0 \leq x < 1$.

Part (i)

With $n = 1$ the equation is $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 1 - x$. Separating (fine, since $1 - x > 0$ along the solution),

$$t = \int_0^x \frac{\mathrm{d}\xi}{1-\xi} = -\ln(1-x), \qquad \text{so} \qquad x = 1 - \mathrm{e}^{-t}.$$

The graph starts at the origin with gradient 1, increases throughout and is concave (its gradient e^{-t} decreases), and approaches the asymptote $x = 1$ as $t \to \infty$ without ever reaching it: $t_1(x) = -\ln(1-x) \to \infty$ as $x \to 1$. The lowest curve in the figure below shows it.

Part (ii)

The inequality. For $0 < x < 1$ both $1 - x$ and $(1 - x^2)^{1/2}$ are positive, so the inequality is equivalent to the one obtained by squaring:

$$(1-x)^2 < 1 - x^2 \iff 1 - 2x + x^2 < 1 - x^2 \iff 2x^2 < 2x \iff x < 1,$$

where the last step divides by $2x > 0$. Since $x < 1$ is true, so is $1 - x < (1 - x^2)^{1/2}$.

The sketch. At every height x in $(0, 1)$ the $n = 2$ curve is steeper than the $n = 1$ curve. Hence, for $0 < x < 1$,

$$t_2(x) = \int_0^x \frac{\mathrm{d}\xi}{(1-\xi^2)^{1/2}} < \int_0^x \frac{\mathrm{d}\xi}{1-\xi} = t_1(x),$$

because the integrand on the left is strictly smaller on $(0, x]$. So the $n = 2$ solution reaches every height strictly sooner, and since both solutions are increasing, at every time $t > 0$ (while $x_2 < 1$) the $n = 2$ curve lies strictly above the $n = 1$ curve. Both leave the origin with gradient 1, so near the origin they are very close, and they separate as t grows.

> **Ask yourself** The two curves have the same gradient at the origin. Why can't the $n = 1$ curve sneak above later?

To get above, it would have to catch up with the $n = 2$ curve at some height x, that is, reach x at the same time. But $t_2(x) < t_1(x)$ for every $x \in (0, 1)$: the slower climber is always strictly late. (Equivalently: wherever the curves are at the same height, the $n = 2$ one is steeper, so it can never be overtaken.)

Solving. Put $x = \sin y$ with $0 \le y < \frac{1}{2}\pi$, and $y = 0$ at $t = 0$. Then $\cos y > 0$, so $(1 - x^2)^{1/2} = (\cos^2 y)^{1/2} = \cos y$, and the equation becomes

$$\cos y \frac{\mathrm{d}y}{\mathrm{d}t} = \cos y \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 1 \quad \Longrightarrow \quad y = t,$$

using $y = 0$ at $t = 0$. So the solution is

$$x = \sin t \qquad \text{for } 0 \le t < \tfrac{1}{2}\pi.$$

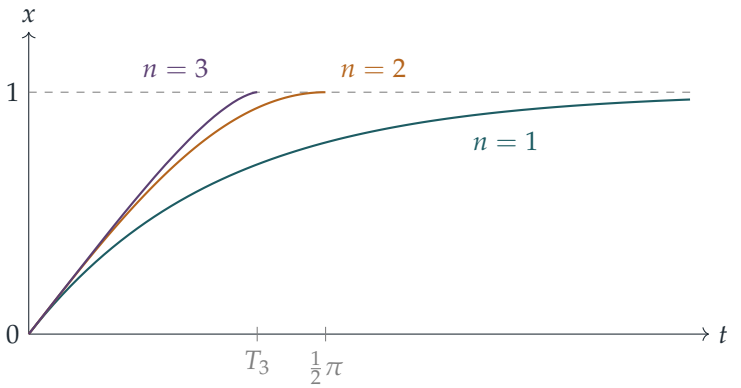
This agrees with $t_2(x) = \arcsin x$. Unlike the case $n = 1$, the solution reaches $x = 1$ in finite time, at $t = \frac{1}{2}\pi$, and arrives there with gradient $\cos \frac{1}{2}\pi = 0$, touching the line $x = 1$.

Part (iii)

Using the given inequality $(1 - x^2)^{1/2} < (1 - x^3)^{1/3}$ on $(0, 1)$ exactly as before, the $n = 3$ curve is steeper than the $n = 2$ curve at every height, so

$$t_3(x) = \int_0^x \frac{\mathrm{d}\xi}{(1-\xi^3)^{1/3}} < \int_0^x \frac{\mathrm{d}\xi}{(1-\xi^2)^{1/2}} = \arcsin x \qquad (0 < x < 1).$$

So the $n = 3$ curve leaves the origin with gradient 1 like the others and lies strictly above $x = \sin t$ for $t > 0$. Its gradient $(1 - x^3)^{1/3}$ falls to 0 as $x \to 1$, so it also flattens out as it meets the line $x = 1$.



Reaching $x = 1$ before $t = \frac{1}{2}\pi$. The sketch shows the $n = 3$ curve above $\sin t$, which itself reaches 1 at $\frac{1}{2}\pi$; so the $n = 3$ curve must get to 1 no later than that. To see that it is strictly earlier, let $x \to 1$ in the comparison of integrals. The time to reach $x = 1$ is

$$T_3 = \int_0^1 \frac{\mathrm{d}\xi}{(1-\xi^3)^{1/3}}.$$

This improper integral converges: since $1 - \xi^3 = (1-\xi)(1+\xi+\xi^2) \geq 1 - \xi$ on $[0,1]$, the integrand is at most $(1-\xi)^{-1/3}$, whose integral over $[0,1]$ is $\frac{3}{2}$. And since the integrand is strictly less than $(1-\xi^2)^{-1/2}$ on $(0,1)$,

$$T_3 < \int_0^1 \frac{\mathrm{d}\xi}{(1-\xi^2)^{1/2}} = \Big[\arcsin\xi\Big]_0^1 = \tfrac{1}{2}\pi.$$

So the $n = 3$ solution reaches $x = 1$ at a time $T_3 < \frac{1}{2}\pi$, as required. (Once there it stays at $x = 1$, since the gradient is 0 and cannot become negative.)

Checks

- Substituting back: $\frac{\mathrm{d}}{\mathrm{d}t}(1 - \mathrm{e}^{-t}) = \mathrm{e}^{-t} = 1 - x$, and $\frac{\mathrm{d}}{\mathrm{d}t}\sin t = \cos t = (1 - \sin^2 t)^{1/2}$ for $0 \leq t \leq \frac{1}{2}\pi$. A numerical (Runge–Kutta) solution of each equation agrees with these formulae.
- Numerically, $T_3 \approx 1.2092$, comfortably less than $\frac{1}{2}\pi \approx 1.5708$. In fact $T_3 = \dfrac{2\pi}{3\sqrt{3}}$ exactly (see below), which matches to seven decimal places.
- At height $x = 0.9$ the three times are $-\ln 0.1 \approx 2.30$, $\arcsin 0.9 \approx 1.12$ and $t_3(0.9) \approx 0.98$, in the predicted order.

Common slips

- Arguing "the gradient is bigger, so the curve is higher" without noting that the comparison is of gradients *at the same height*, not at the same time. Turning to t as a function of x makes the argument honest.

- Writing $(1-\sin^2 y)^{1/2} = \cos y$ without saying why $\cos y \geq 0$: it needs $0 \leq y \leq \frac{1}{2}\pi$.
- Continuing $x = \sin t$ past $t = \frac{1}{2}\pi$: there it would decrease, but the equation forces $\mathrm{d}x/\mathrm{d}t \geq 0$.

Beyond STEP

The arrival at $x = 1$ in finite time, with the solution then sitting at $x = 1$ for ever, happens because $(1-x^n)^{1/n}$ is not Lipschitz at $x = 1$; this is exactly where the uniqueness theorem for differential equations breaks down (for $n = 2$, both $x = 1$ and $x = \sin t$ pass through $(\frac{1}{2}\pi, 1)$). The arrival time is a Beta function: substituting $u = \xi^n$ gives $T_n = \frac{1}{n}B(\frac{1}{n}, 1-\frac{1}{n}) = \dfrac{\pi/n}{\sin(\pi/n)}$, which is $\frac{1}{2}\pi$ for $n = 2$, $\frac{2\pi}{3\sqrt{3}}$ for $n = 3$, and decreases towards 1 as $n \to \infty$. Boyce and DiPrima, *Elementary Differential Equations*, discusses existence and uniqueness of solutions and comparison of solutions.

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